

SpaceX Falcon 9 First Stage Landing Prediction

GitHub Repository:

<https://github.com/Satish-KumarRam/IBM-Data-Science-Capstone-Project>

**IBM Data Science Professional Certificate
Capstone Project**

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3 August 2026

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Introduction

- **Project Background and Context:**

- SpaceX has significantly reduced the cost of space missions by successfully recovering and reusing Falcon 9 first-stage boosters. Traditional rocket launches are expensive because rockets are usually discarded after launch. By predicting whether the Falcon 9 first stage will land successfully, organizations can estimate mission costs, improve planning, and support data-driven decision-making. This project uses historical SpaceX launch data to analyze landing outcomes and build machine learning models for landing prediction.

- **Project Objective:**

- The project aims to identify the factors influencing Falcon 9 first-stage landing success and develop a machine learning model to predict landing outcomes.
- Collect Falcon 9 launch data using the SpaceX REST API and **Wikipedia Web Scraping**.
- Perform **data wrangling, SQL analysis, and exploratory data analysis (EDA)**.
- Build interactive visualizations using SQL, Folium, and Plotly Dash.
- Develop, evaluate, and compare machine learning classification models to predict Falcon 9 first-stage landing success.

Executive Summary

- **Objective**

- Predict Falcon 9 first-stage landing success using historical launch data and machine learning.

- **Summary of Methodology**

- Data Collection through **SpaceX REST API**.
- Data Collection using **Wikipedia Web Scraping**.
- Data Wrangling and Feature Engineering.
- Exploratory Data Analysis (EDA) using SQL Queries, Python Visualizations (Matplotlib & Seaborn), and Plotly Dash.
- Interactive Visual Analytics using **Folium** and **Plotly Dash**.
- Predictive Analysis using **Classification Models**.

- **Summary of Results**

- Landing success rate improved significantly over time.
- Payload mass, orbit type, and launch site influenced landing success.
- SQL and visualizations revealed important launch patterns.
- Interactive dashboards provided deeper insights into launch data.
- The best classification model achieved the highest prediction accuracy.

Methodology

- **Data Collection Methodology**

- Collected Falcon 9 launch data using the **SpaceX REST API**.
- Collected additional launch records through **Wikipedia Web Scraping**.

- **Data Wrangling**

- Cleaned missing values and standardized the dataset.
- Created the landing success label and prepared features for analysis.

- **Exploratory Data Analysis (EDA)**

- Performed **SQL queries** to analyze launch statistics.
- Created visualizations using Matplotlib and Seaborn.

- **Interactive Visual Analytics**

- Built an interactive **Folium** map to analyze launch site locations.
- Developed an interactive Plotly Dash dashboard with dropdown filters, charts, and data visualizations for launch analysis.

- **Predictive Analysis**

- Built, trained, and evaluated multiple machine learning classification models.
- Compared model performance and selected the best-performing model based on prediction accuracy.



Data Collection using SpaceX API

- Collected Falcon 9 launch data using the **SpaceX REST API**.
- Retrieved launch date, payload mass, orbit, launch site, booster version, and mission outcome.
- Converted JSON responses into a structured Pandas DataFrame for analysis.
- **Right side:** SpaceX API notebook screenshot.

```
import requests
import pandas as pd

static_json_url = "https://cf-courses-data.s3.us.cloud-object-storage.appdomain.cloud/IBM-DS0321EN-SkillsNetwork/datasets/API_call_spacex_api.json"

response = requests.get(static_json_url)

print(response.status_code)

200

We should see that the request was successful with the 200 status response code

response.status_code

200

Now we decode the response content as a Json using .json() and turn it into a Pandas dataframe using .json_normalize()

# Use json_normalize method to convert the json result into a dataframe
data = pd.json_normalize(response.json())

Using the dataframe data print the first 5 rows

# Get the head of the dataframe
data.head(5)
```

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Data Collection using Wikipedia Web Scrapping

- Extracted additional Falcon 9 launch records from **Wikipedia** using **BeautifulSoup**.
- Parsed HTML tables and converted them into a Pandas DataFrame.
- Used the scraped data to enrich and validate the dataset.
- Successfully extracted Falcon 9 launch tables using BeautifulSoup for further analysis.

```
# use requests.get() method with the provided static_url and headers
# assign the response to a object
response = requests.get(static_url, headers=headers)
```

Create a `BeautifulSoup` object from the HTML `response`

```
# Use BeautifulSoup() to create a BeautifulSoup object from a response text content
soup = BeautifulSoup(response.text, "html.parser")
```

```
print(soup.title.string)
```

List of Falcon 9 and Falcon Heavy launches - Wikipedia

Print the page title to verify if the `BeautifulSoup` object was created properly

```
# Use soup.title attribute
print(soup.title)
```

<title>List of Falcon 9 and Falcon Heavy launches - Wikipedia</title>

```
# Use the find_all function in the BeautifulSoup object, with element type `table`
# Assign the result to a list called `html_tables`
```

```
html_tables = soup.find_all('table')
```

Starting from the third table is our target table contains the actual launch records.

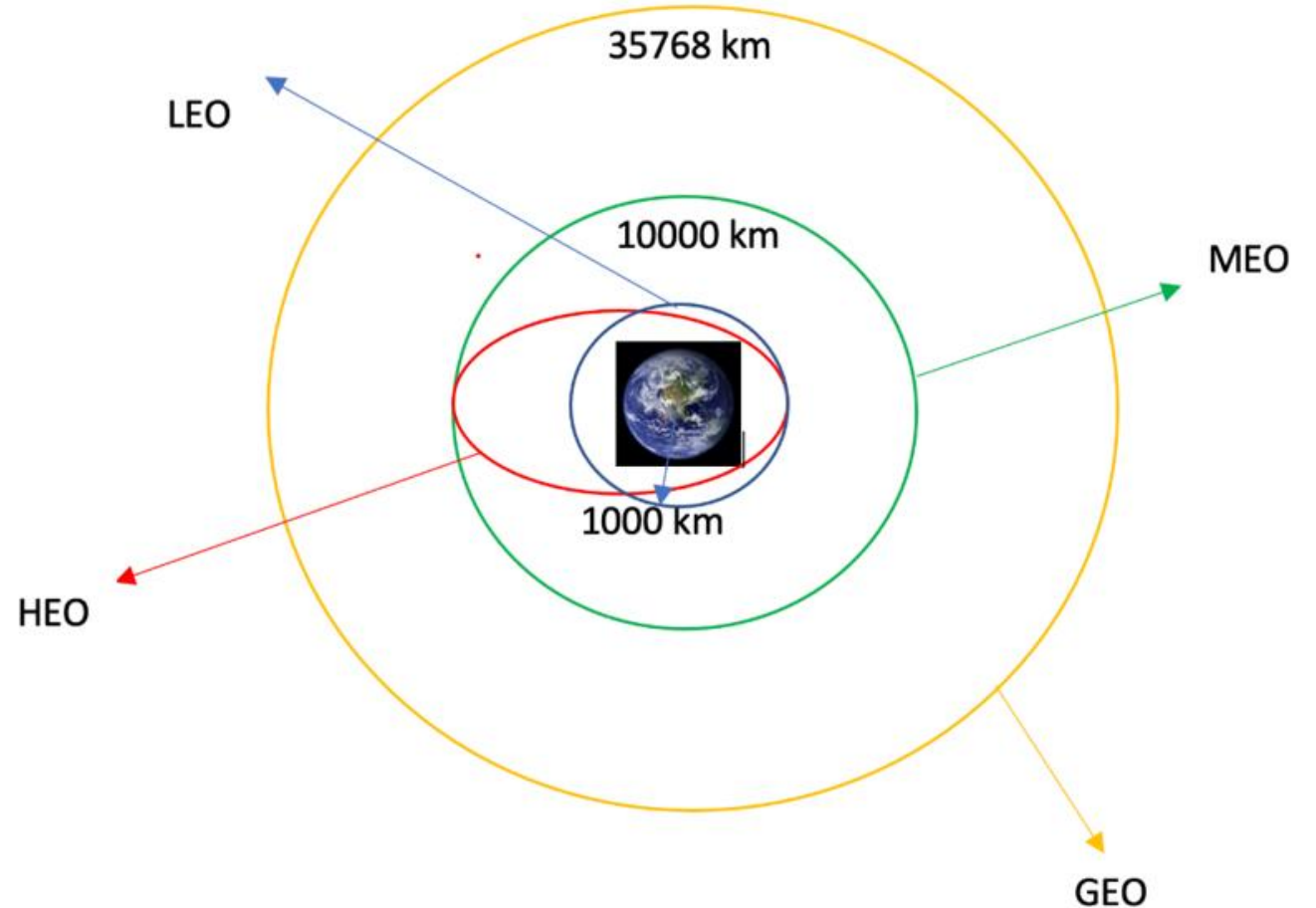
```
# Let's print the third table and check its content
first_launch_table = html_tables[2]
print(first_launch_table)
print(first_launch_table)
```


Data Wrangling

- Cleaned missing and inconsistent values.
- Selected relevant features for analysis.
- Created the landing success label.
- Prepared the final dataset for EDA and machine learning.

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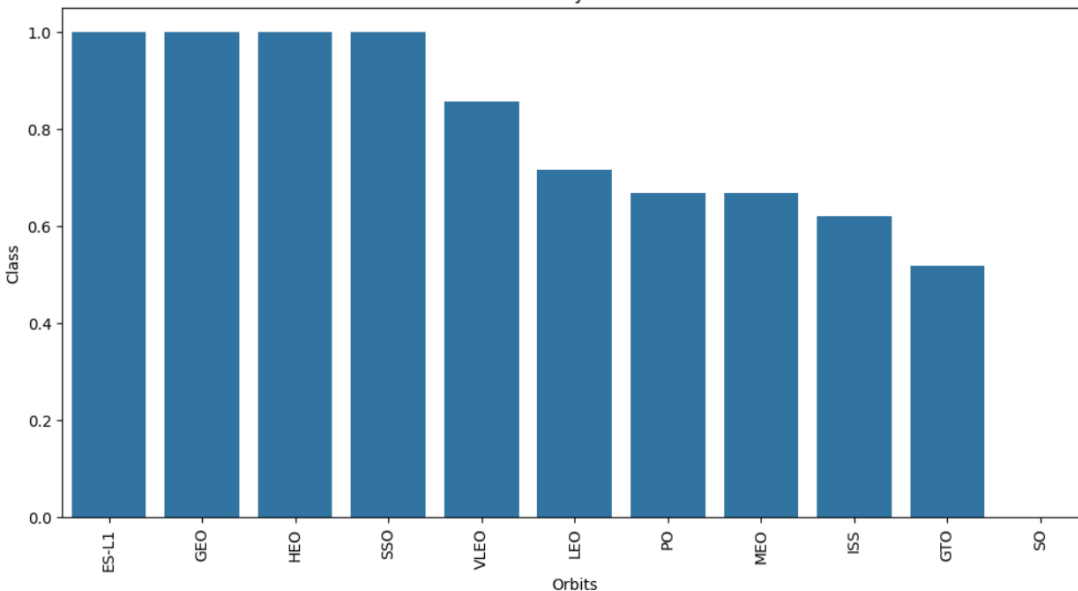
<https://github.com/Satish-KumarRam/IBM-Data-Science-Capstone-Project>



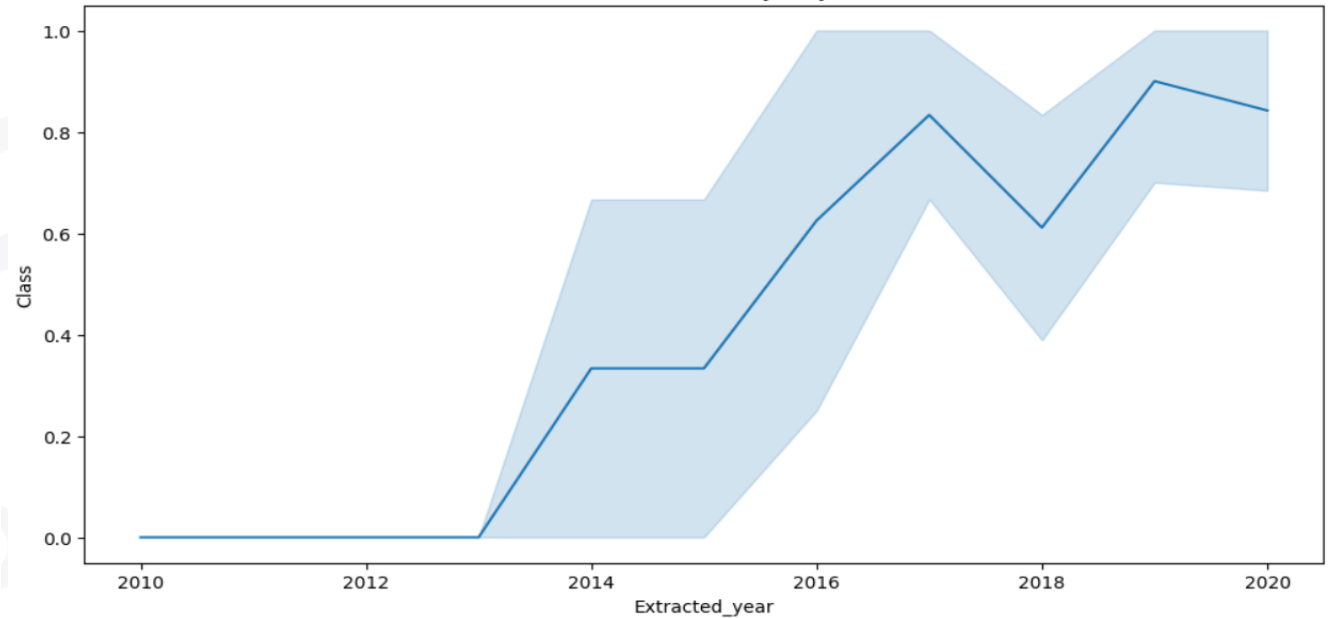
EDA using Visualizations

- Launch success increased over time.
- Success rates differed across orbit classes.
- EDA revealed patterns influencing landing success.

Plot of success rate by class of each Orbits



Plot of launch success yearly trend



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<https://github.com/Satish-KumarRam/IBM-Data-Science-Capstone-Project>



EDA using SQL

- Used SQL queries to summarize launch statistics.
- Analyzed launches by launch site, mission outcome, and booster version.
- Identified trends and generated insights from the data.

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EDA Using SQL Queries (Part 1)

SQL Query Output – Launch Site Analysis

```
%%sql
```

```
SELECT Launch_Site,  
COUNT(*) AS Total_Launches  
FROM SPACEXTABLE  
GROUP BY Launch_Site;
```

```
* sqlite:///my_data1.db  
Done.
```

Launch_Site	Total_Launches
CCAFS LC-40	26
CCAFS SLC-40	34
KSC LC-39A	25
VAFB SLC-4E	16

- **Key Findings:**
- Used SQL to calculate total launches for each Falcon 9 launch site.
- **CCAFS SLC-40** recorded the highest number of launches (**34**).
- **CCAFS LC-40** completed **26** launches during the analyzed period.
- KSC LC-39A completed 25 launches.
- **VAFB SLC-4E** recorded the fewest launches (**16**).
- These SQL results identified the busiest launch sites and supported subsequent visualization and predictive analysis.

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EDA Using SQL Queries (Part 2)

Mission Outcome Analysis (SQL Query Result)

```
%%sql

SELECT Mission_Outcome,
       COUNT(*) AS Count
FROM SPACEXTABLE
WHERE Date BETWEEN '2010-06-04' AND '2017-03-20'
GROUP BY Mission_Outcome
ORDER BY Count DESC;
```

* sqlite:///my_data1.db

Done.

Mission_Outcome	Count
Success	30
Failure (in flight)	1

- **Key Findings:**
- SQL was used to summarize mission outcomes.
- **30 missions** were completed successfully.
- Only **1 mission** resulted in failure during the selected period.
- The high success rate indicates improved Falcon 9 reliability.
- SQL aggregation helped validate mission performance before predictive modelling.
- These SQL results supported exploratory data analysis and feature selection for machine learning.

EDA Using SQL Queries (Part 3)

```
[18]: %%sql
      SELECT Booster_Version
      FROM SPACEXTABLE
      WHERE Payload_Mass__kg_ BETWEEN 4000 AND 6000;

* sqlite:///my_data1.db
Done.
```

[18]: **Booster_Version**

F9 v1.1
F9 v1.1 B1011
F9 v1.1 B1014
F9 v1.1 B1016
F9 FT B1020
F9 FT B1022
F9 FT B1026
F9 FT B1030
F9 FT B1021.2
F9 FT B1032.1
F9 B4 B1040.1

- **Key Findings:**

- SQL was used to identify booster versions carrying payloads between **4000 kg and 6000 kg**.
- Multiple Falcon 9 booster versions operated within this payload range.
- Medium-weight payload missions were common across several booster versions.
- These findings support payload trend analysis and provide useful insights for predictive modelling.

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Interactive Visual Analytics (Folium & Plotly Dash)

- Built interactive **Folium** maps to visualize launch site locations.
- Developed a **Plotly Dash** dashboard with filters and interactive charts.
- Enabled dynamic exploration of launch data.

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Predictive Analysis

- Built multiple classification models to predict landing success.
- Compared model performance using evaluation metrics.
- Selected the best-performing model based on prediction accuracy.

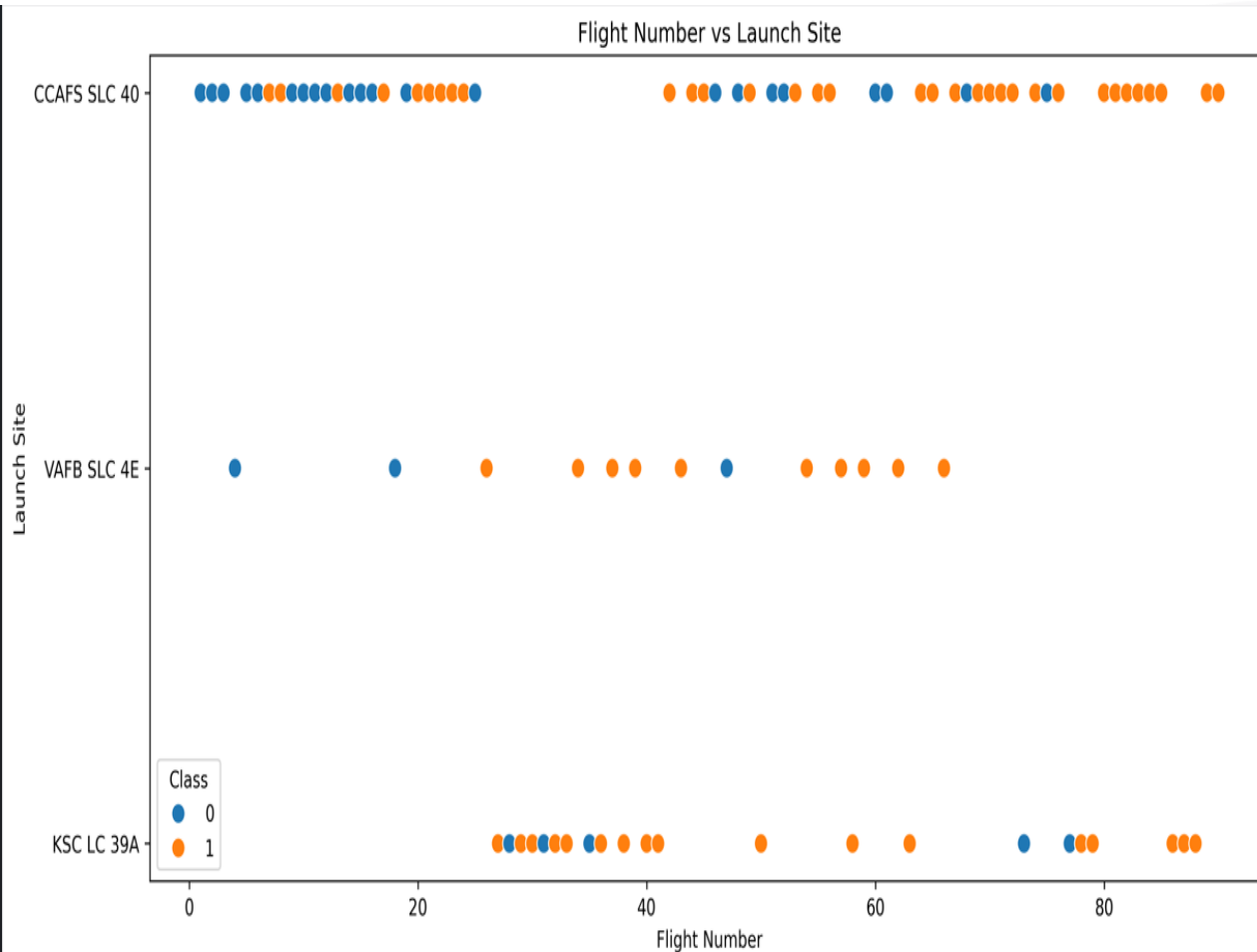
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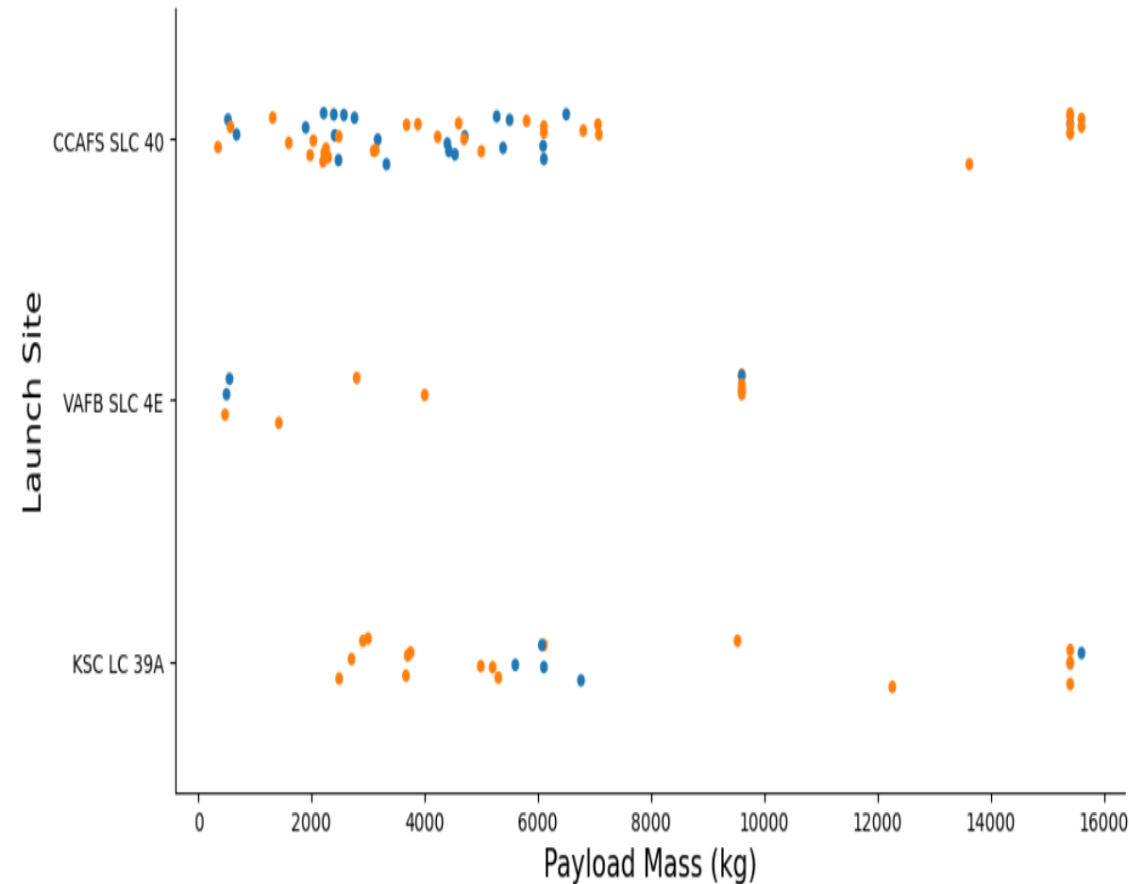
Results

- **Landing Success Trend:** This project integrates data collection, EDA, SQL, interactive dashboards, and machine learning to predict Falcon 9 landing success.
- **Key Factors:** Payload mass, launch site, orbit type, flight number, and booster version influenced landing success.
- **EDA, SQL & Visualization:** SQL queries, Folium maps, and the Plotly Dash dashboard provided interactive insights.
- **Predictive Analysis:** The best classification model achieved the highest prediction accuracy.
- **Overall Outcome:** Historical launch data can effectively predict Falcon 9 first-stage landing success and support mission planning.

Flight Number vs. Launch Site



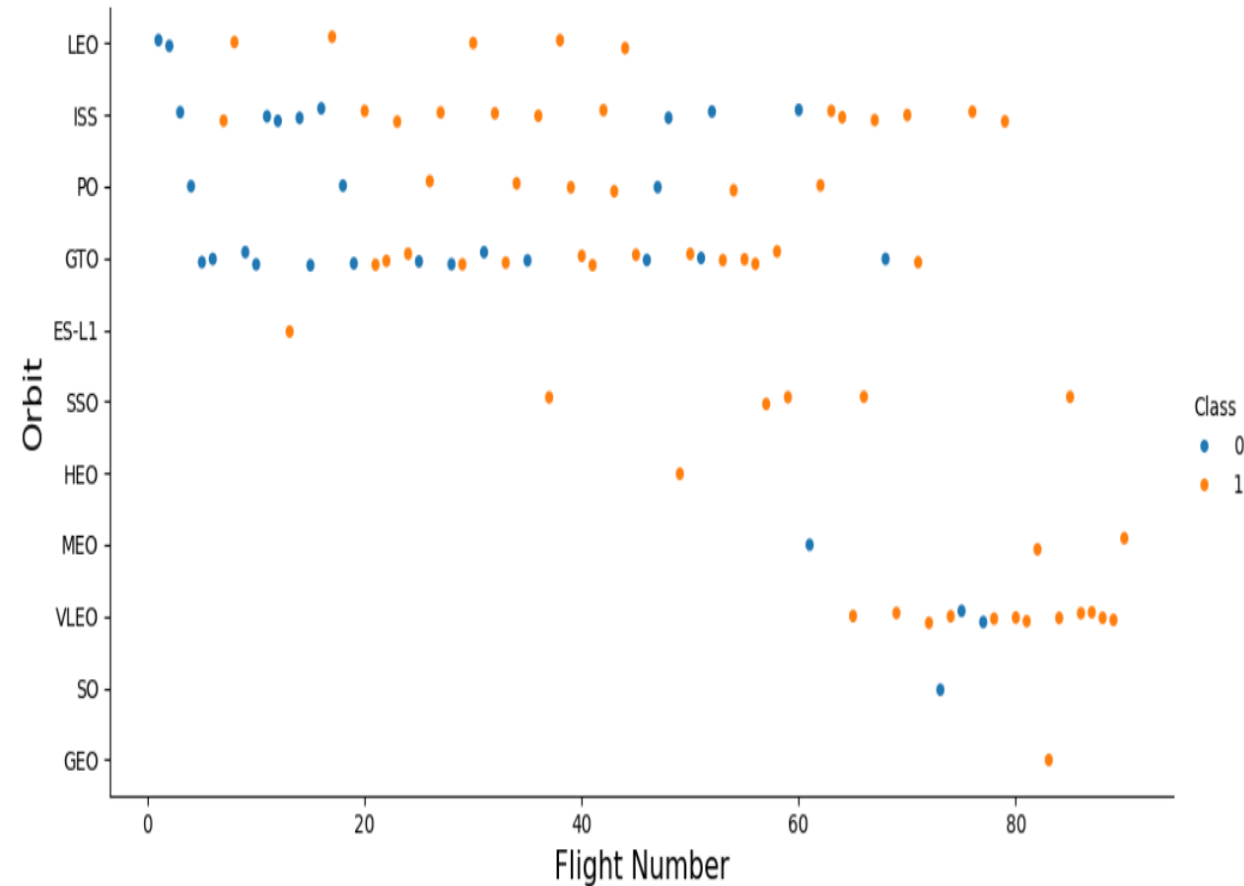
Payload Mass vs. Launch Site



- **Key Findings**

- **CCAFS SLC 40** handled the widest range of payload masses and was the most frequently used launch site.
- **KSC LC 39A** primarily handled medium- to high-payload missions and achieved several successful landings.
- **VAFB SLC 4E** conducted fewer launches but demonstrated successful missions across different payload categories.
- Both **successful** and **unsuccessful** landings occurred at low, medium, and high payload masses, indicating that **payload mass alone was not the primary factor affecting landing success**.
- Overall, launch site and payload mass together influenced Falcon 9 landing success.

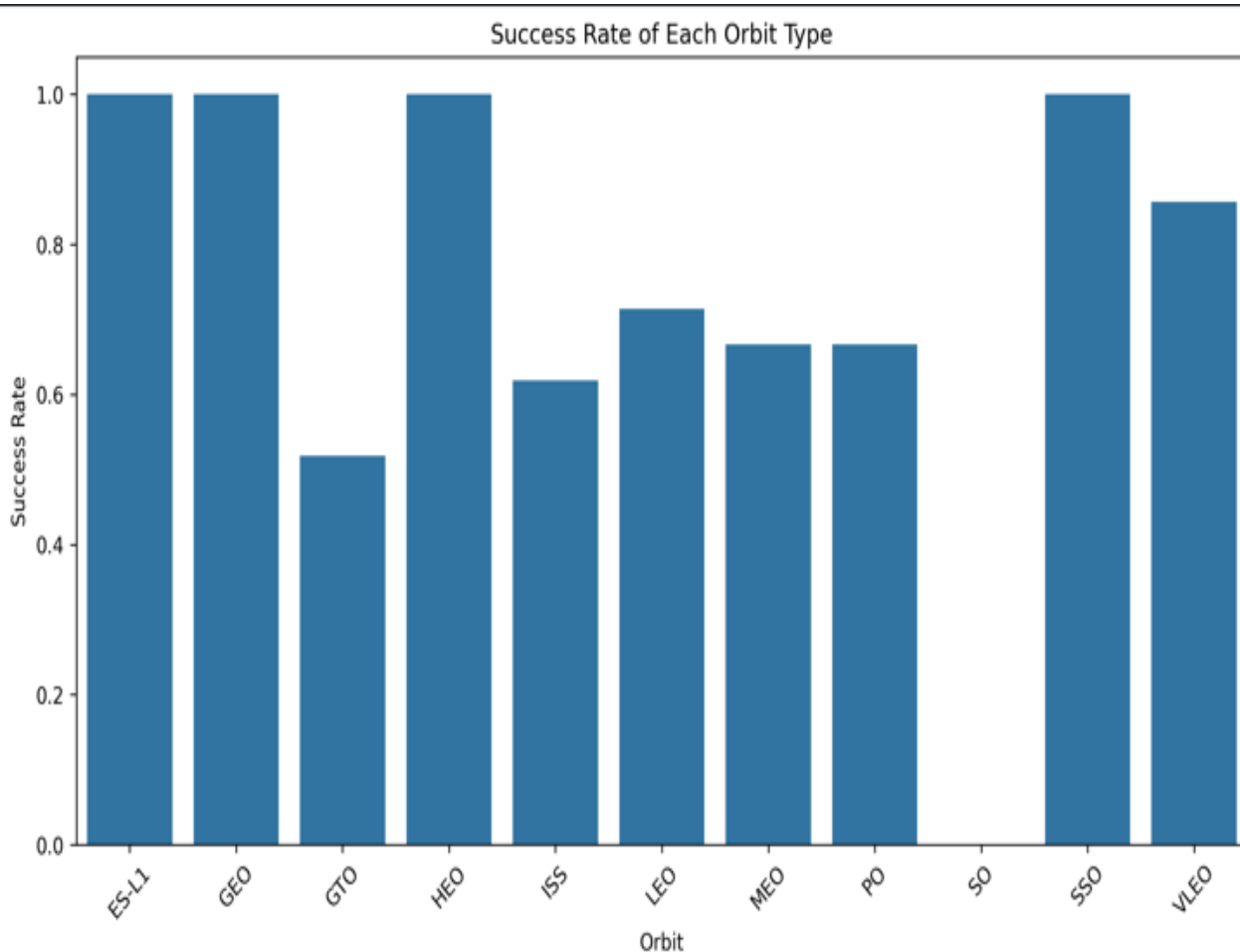
Flight Number vs. Orbit



• Key Findings

- LEO, ISS, and VLEO missions generally achieved high landing success rates.
- GTO missions showed mixed landing outcomes, with both successful and failed landings.
- Landing success generally improved with increasing flight numbers, especially in later missions.
- Orbit type influenced landing success, while payload mass and launch site also played important roles.

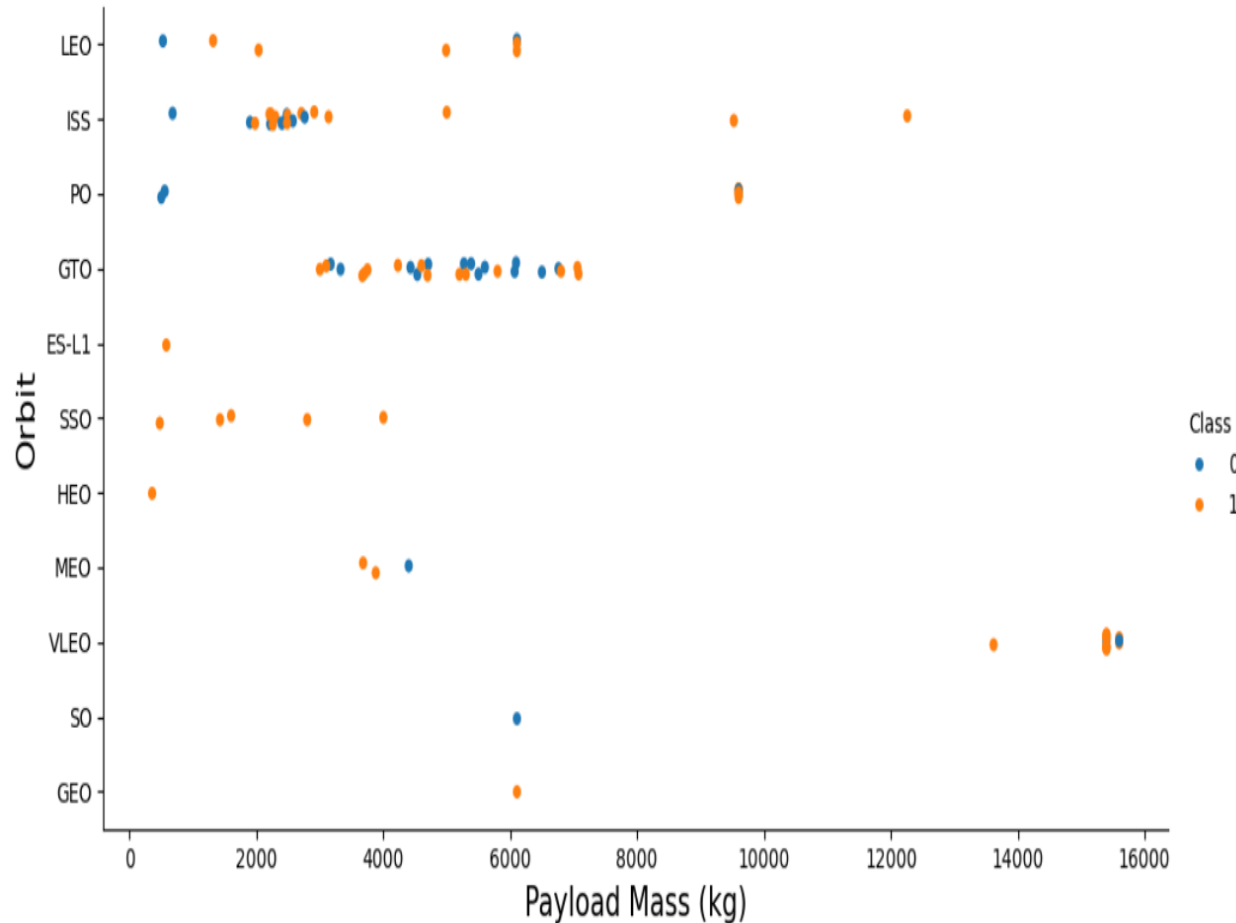
Success Rate by Orbit



Key Findings

- **ES-L1, GEO, HEO, and SSO** achieved a **100% landing success rate**, indicating highly reliable mission outcomes for these orbit types.
- **VLEO** also demonstrated a high landing success rate of approximately **86%**, showing strong booster recovery performance.
- LEO, MEO, ISS, and PO achieved moderate landing success rates, indicating variable mission outcomes across these orbit types.
- **GTO** recorded a comparatively lower landing success rate, while **SO** showed **no successful landings** in the available dataset.
- Overall, orbit type significantly influenced Falcon 9 landing success, although mission characteristics and operational factors also played important roles.

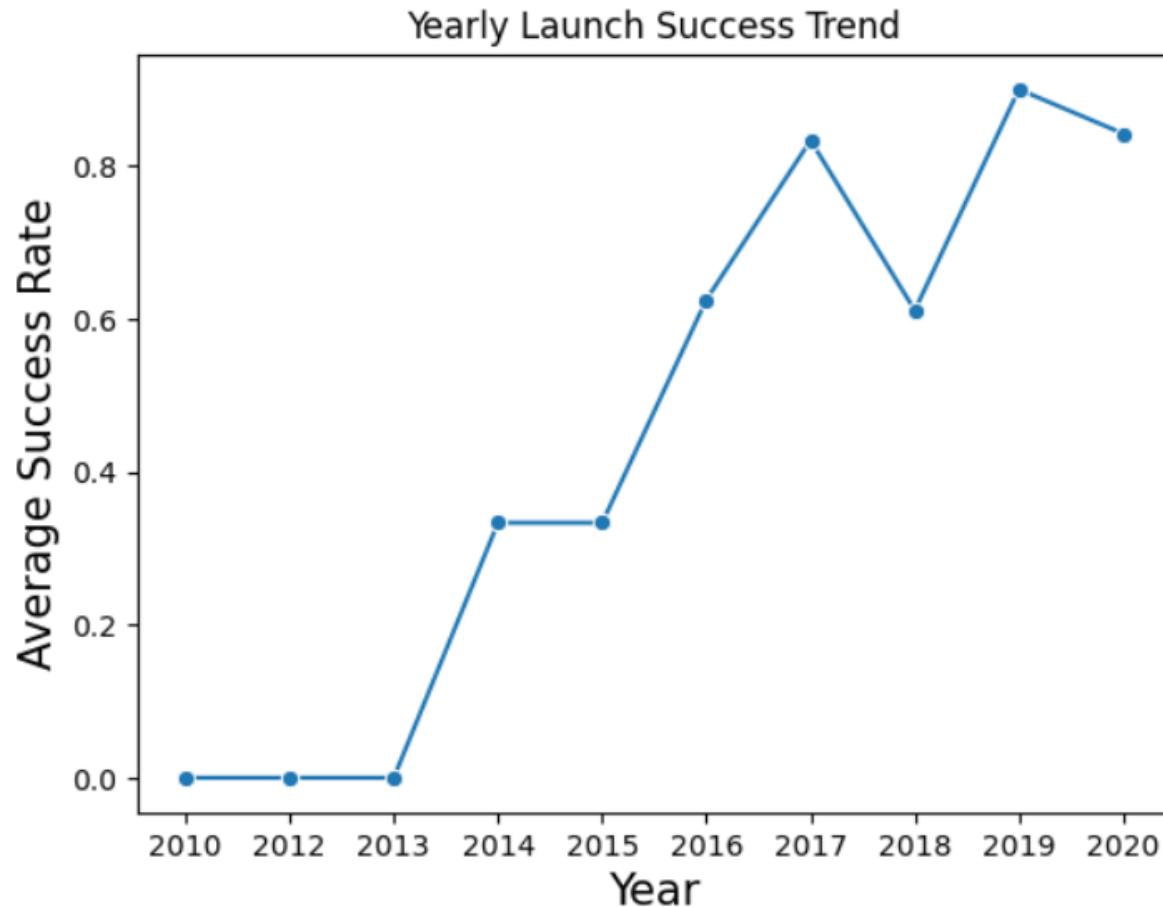
Payload Mass vs. Orbit



• Key Findings

- GTO missions covered a wide range of payload masses, with both successful and unsuccessful landings observed.
- VLEO missions were associated with the highest payload masses.
- LEO and ISS missions generally carried low to medium payload masses.
- Payload mass alone did not determine Falcon 9 landing success; orbit type and mission characteristics also influenced landing outcomes.

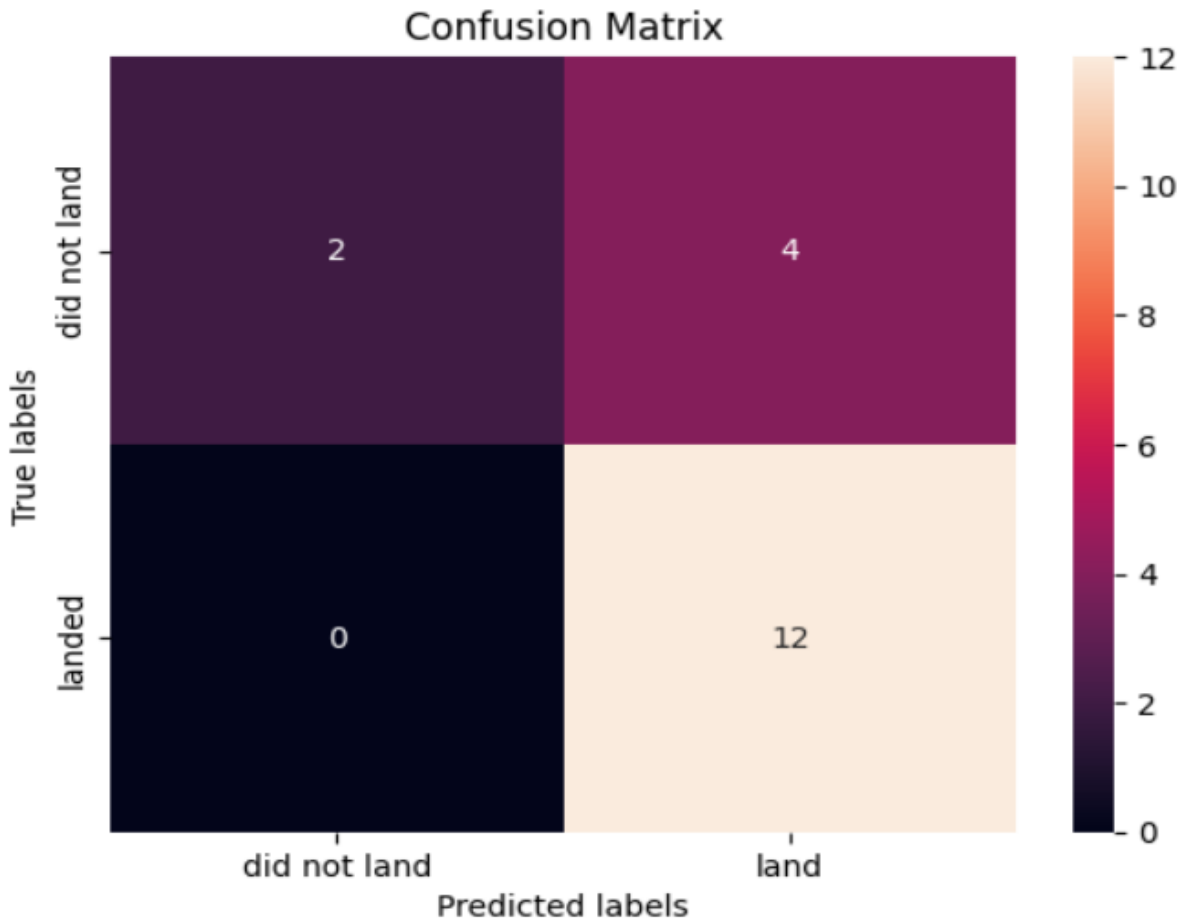
Yearly Launch Success Trend



• Key Findings

- Overall launch success rates improved over the years despite minor fluctuations.
- Success rates were very low during the early years (2010–2013).
- From 2016 onward, average launch success rates increased significantly.
- By 2019–2020, SpaceX consistently achieved high launch success rates, demonstrating improved booster recovery and mission reliability.

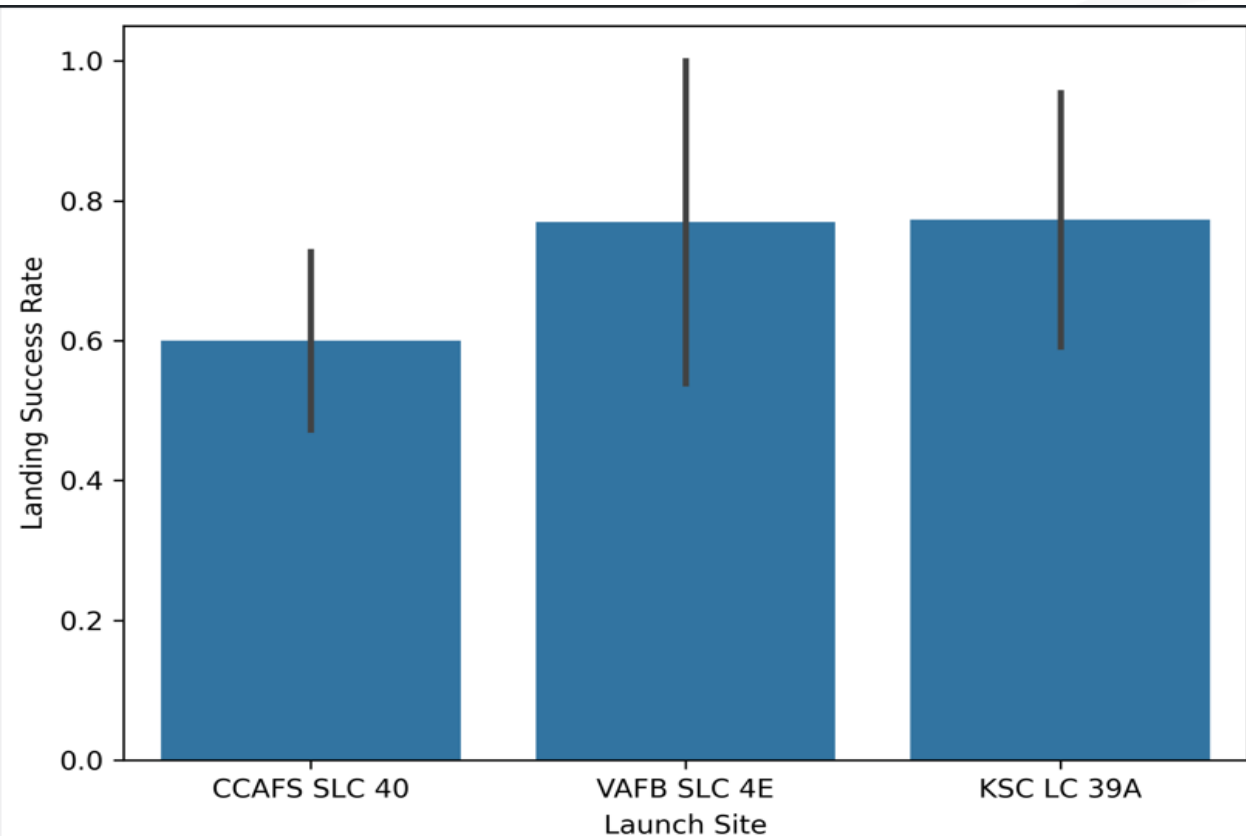
Model Evaluation – Confusion Matrix



- **Key Findings**

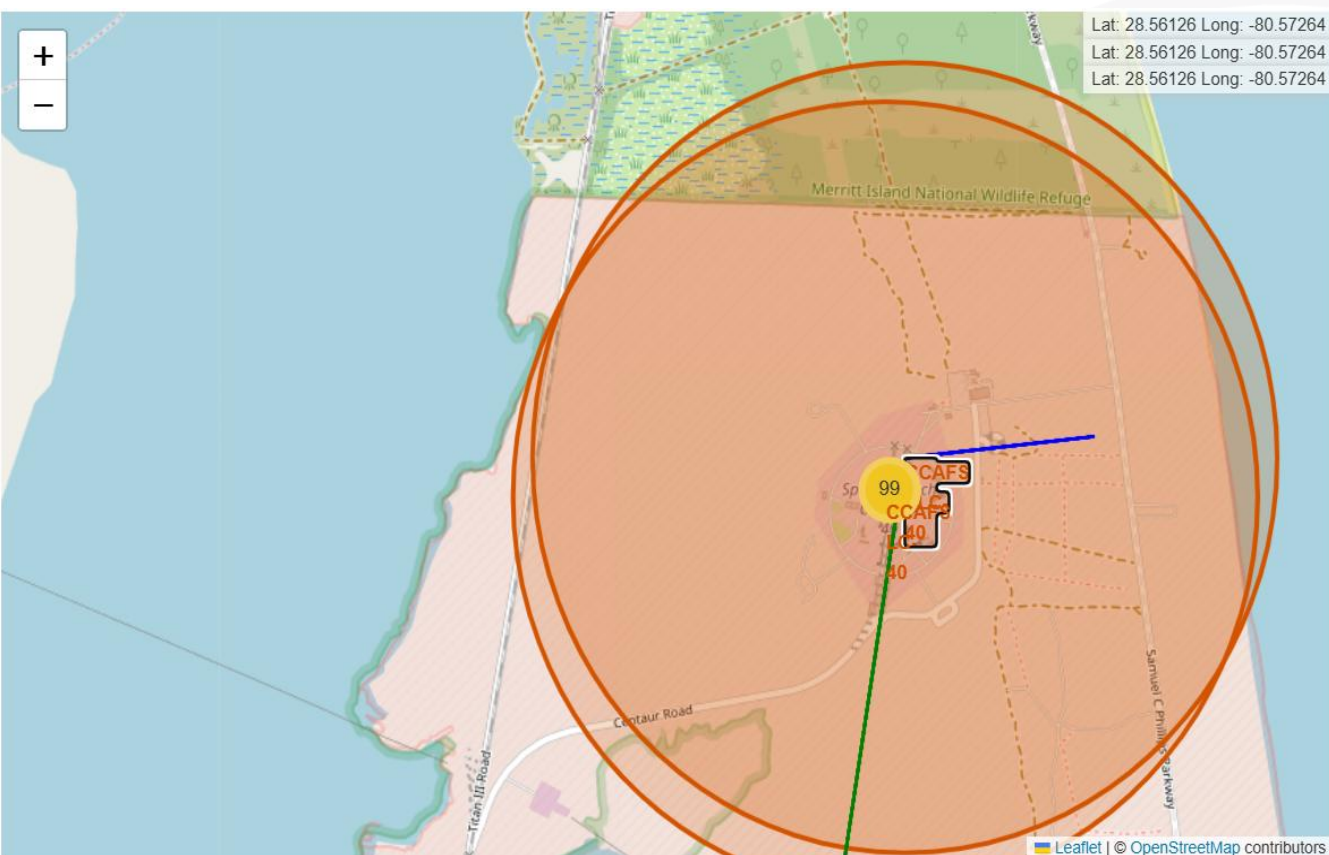
- The model achieved good predictive performance with no false negatives, although a few unsuccessful missions were classified as successful.
- **2 unsuccessful landings** were correctly classified (True Negatives).
- **4 unsuccessful landings** were incorrectly predicted as successful (False Positives).
- The model demonstrated strong performance in identifying successful landings, with no false negatives, although some unsuccessful missions were incorrectly classified as successful.

Landing Success by Launch Site



- **Key Findings**
- **KSC LC 39A** achieved the **highest average landing success rate** among all launch sites.
- **VAFB SLC 4E** also recorded a **high landing success rate**, indicating strong mission performance.
- **CCAFS SLC 40** showed the **lowest average landing success rate** in the dataset.
- Launch site significantly influenced Falcon 9 first-stage landing success.

Launch Site Proximity Analysis (CCAFS)



• Key Findings:

- CCAFS launch site is located close to the coastline.
- The coastal location supports safe launch trajectories over the ocean.
- The site is positioned away from major populated areas to improve safety.
- The proximity map highlights CCAFS's strategic coastal location for safe launch operations.

SPACEX FALCON 9 – FINDINGS & IMPLICATIONS

Findings

- Launch success rate improved over time.
- Launch site, payload mass, flight number, and orbit type significantly influenced landing success.
- The best-performing classification model demonstrated strong predictive performance.

Implications

- Machine learning can effectively predict Falcon 9 landing success.
- The predictive model can support mission planning and reduce launch costs.
- Data-driven insights help improve future SpaceX launch decisions.

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Results – Visualizations & Dashboard

EDA Visualizations

- Flight Number vs. Launch Site
- Payload Mass vs. Launch Site
- Success Rate by Orbit
- Flight Number vs. Orbit
- Payload Mass vs. Orbit
- Yearly Launch Success Trend
- Landing Success by Launch Site

Interactive Dashboard

- SQL Query Results
- Folium Launch Site Map
- Plotly Dash Dashboard
- Confusion Matrix (Model Evaluation)

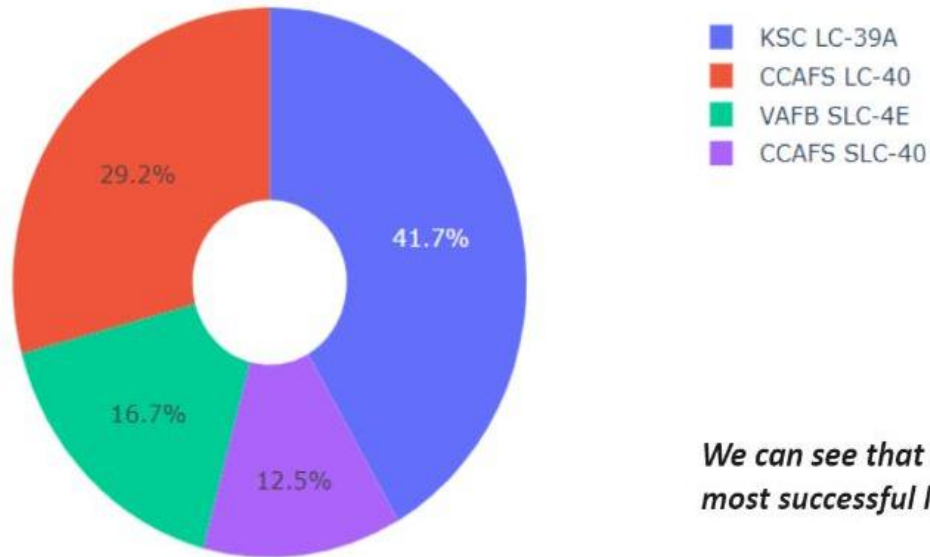
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Interactive Dashboard (Plotly Dash)

Total Success Launches by All Sites (Pie Chart)

Total Success Launches By all sites



We can see that KSC LC-39A had the most successful launches from all the sites

• Key Findings:

- Interactive dashboard developed using Plotly Dash.
- Displays successful launches across all launch sites.
- KSC LC-39A recorded the highest number of successful launches.
- Figure: Plotly Dash dashboard showing the success distribution across all launch sites.

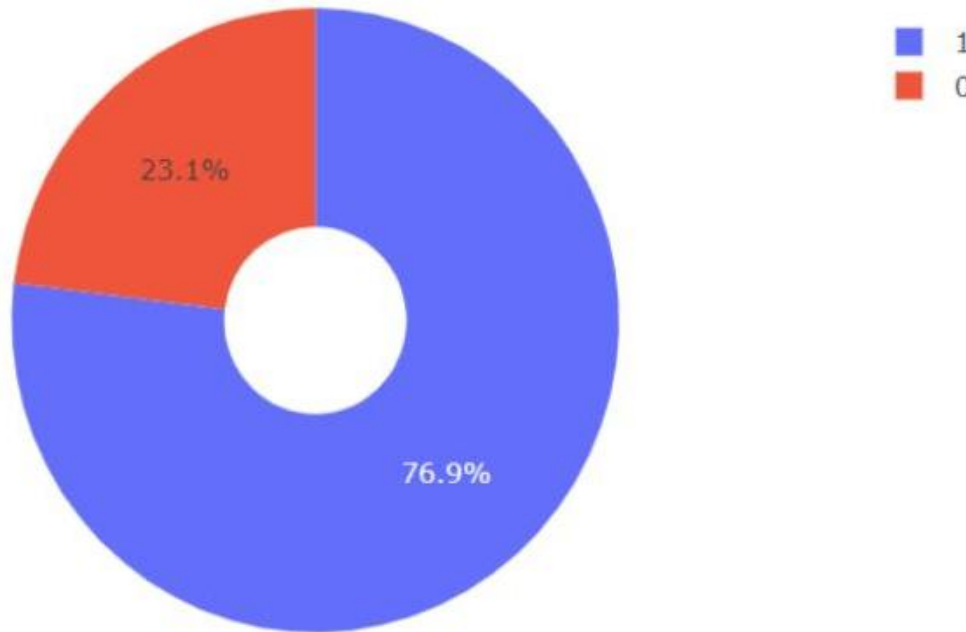
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Launch Site Analysis

KSC LC-39A: Success vs Failure



KSC LC-39A achieved a 76.9% success rate while getting a 23.1% failure rate

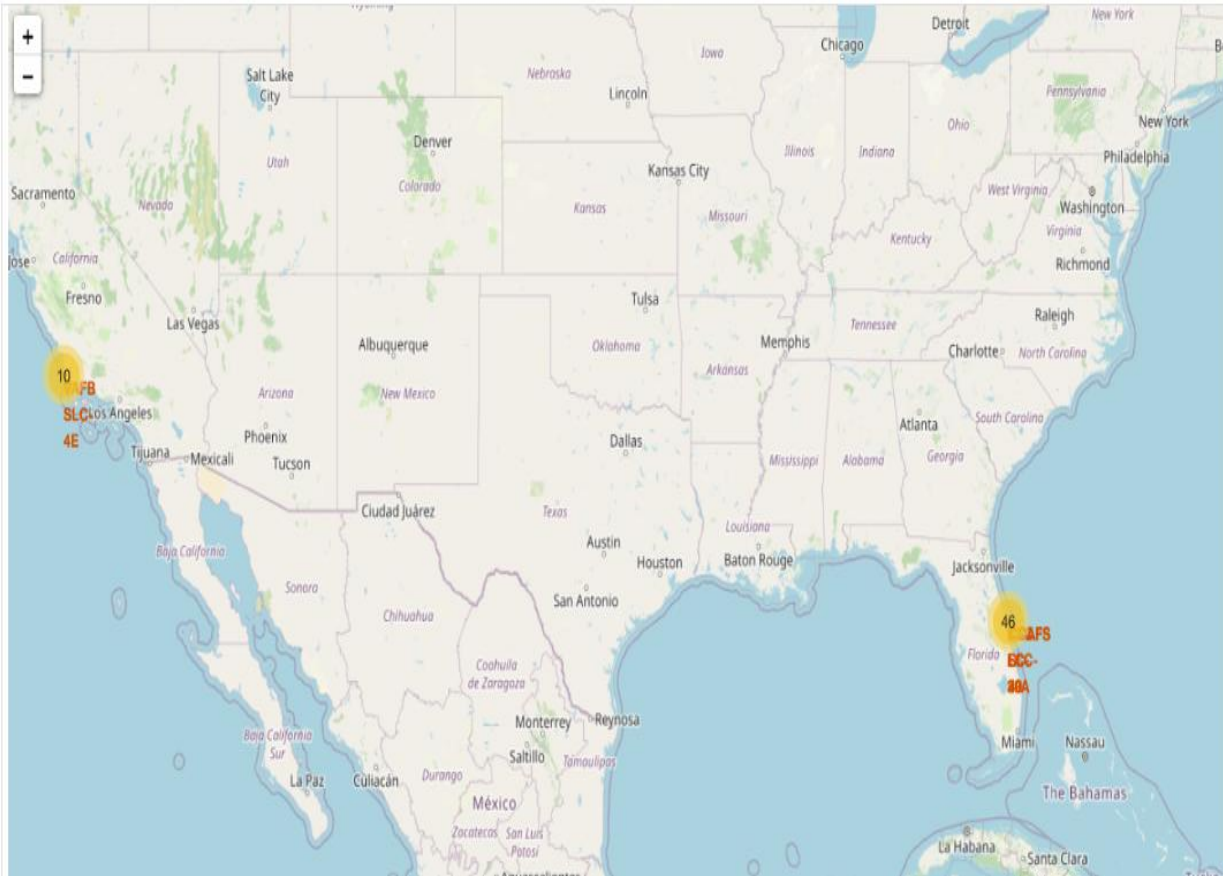
- **Key Findings:**
- Shows the success and failure distribution for **KSC LC-39A**.
- Approximately **76.9%** of launches were successful.
- Helps compare mission outcomes at the selected launch site.
- Demonstrates the effectiveness of Falcon 9 operations at KSC LC-39A.

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Interactive Launch Site Map (Folium)

Launch Site Distribution (Folium)



- **Key Findings:**
- Visualized Falcon 9 launch sites using Folium.
- Interactive markers display launch site locations and launch counts.
- CCAFS SLC-40 recorded the highest number of Falcon 9 launches, indicating it was the most active launch site.
- Geographic analysis helped compare launch site distribution.
- Location-based visualization helped identify launch site patterns and supported predictive analysis.

Interactive Marker Cluster Analysis

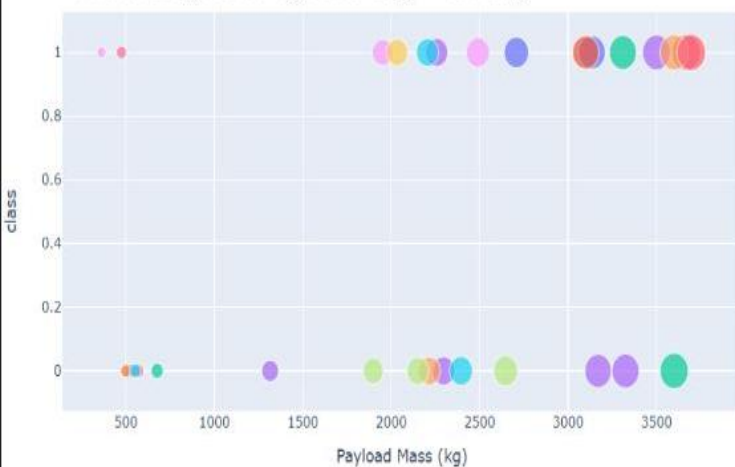
• Key Findings:

- Folium was used to visualize individual Falcon 9 launch records at **CCAFS SLC-40**.
- **Green markers** represent successful launches, while **red markers** represent failed launches.
- Marker clustering improves map readability when multiple launches occur at the same site.
- The visualization supports geographic analysis and helps compare launch outcomes across the launch site.

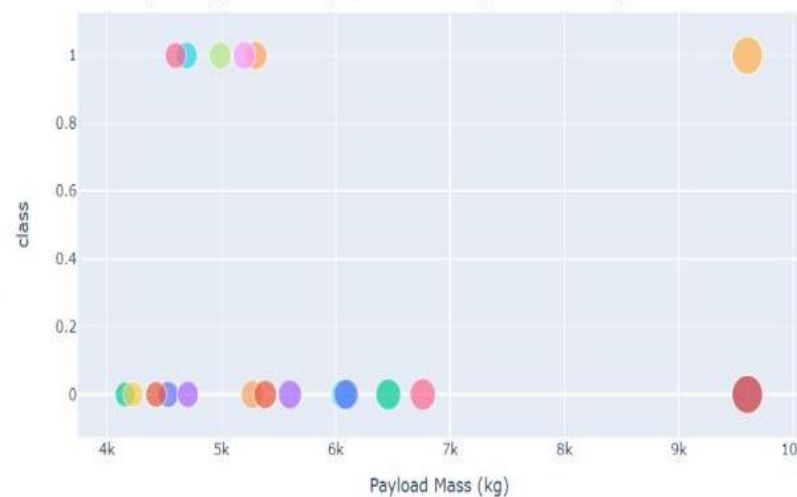
Payload Analysis

Payload vs Launch Success

Low Weighted Payload 0kg – 4000kg



Heavy Weighted Payload 4000kg – 10000kg



We can see the success rates for low weighted payloads is higher than the heavy weighted payloads

- **Key Findings:**
- Payload mass influences launch success differently across booster versions.
- Lower payloads generally show higher success rates.
- Scatter plot helps identify relationships between payload, booster version, and landing outcome.

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Conclusion

• Key Takeaways

- The analysis identified several factors that influence SpaceX Falcon 9 landing success.
- Flight Number showed a positive relationship with landing success.
- Payload mass, orbit type, and launch site were key factors influencing landing success.
- The project successfully combined **SQL, Python, Plotly Dash, and Folium** to analyze Falcon 9 launch data and support predictive insights.
- **Future Work:** Incorporate weather conditions and real-time launch data to further improve prediction accuracy.

• Overall Findings

- Landing success rates improved significantly over time, reaching their highest levels by 2019 and remaining consistently high thereafter.
- Historical launch data provides valuable insights into Falcon 9 landing success and supports data-driven decision-making.
- The Random Forest classifier achieved the best prediction performance among the evaluated machine learning models.
- Combining SQL analysis, EDA, Folium mapping, Plotly Dash, and machine learning provided a comprehensive understanding of Falcon 9 landing success.

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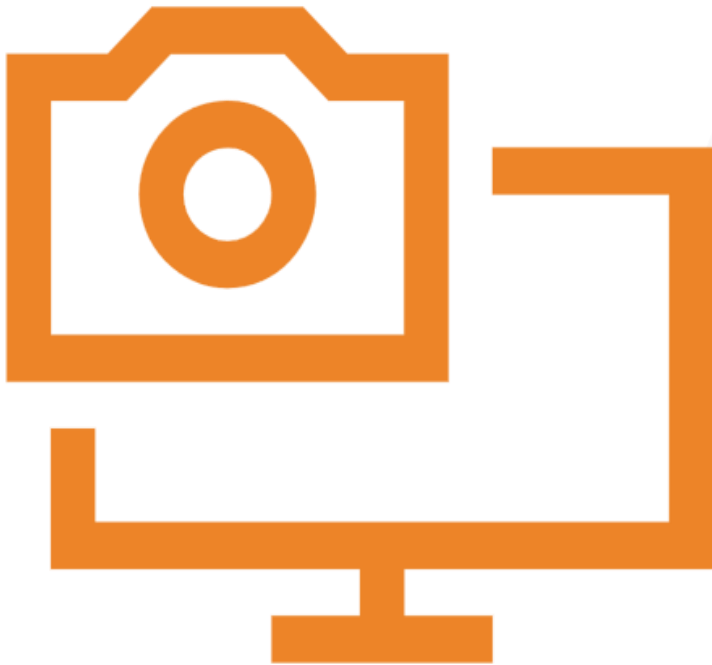
APPENDIX

- **Additional Analysis**

- SQL query outputs used for exploratory analysis.
- Interactive Plotly Dash dashboard for exploratory data analysis.
- Folium map showing launch site proximity.
- Additional EDA charts supporting model development.

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Thank You!



Questions?

